

SPACE RACE



GITHUB LINK

- http://github.com/haduongphan123/IBM_Learning/tree/main

EXECUTIVE SUMMARY

Summary of methodologies

- 1. Data Acquisition and Integration** (SpaceX API-based data extraction and web scraping from Wikipedia)
- 2. Data Cleaning, Transformation, and Database Management** (Data preprocessing, formatting, Db2 database storage, and SQL-based data querying)
- 3. Feature Engineering and Data Standardization**
- 4. Interactive Visualization and Dashboard Development** (Geospatial mapping with Folium and interactive analytics dashboard using Plotly Dash)
- 5. Machine Learning Model Development and Hyperparameter Optimization** (Implementation of SVM, Decision Tree, and KNN models with GridSearchCV tuning)
- 6. Model Performance Evaluation and Comparative Analysis** (Accuracy assessment on test data and model performance comparison)

Summary of result

1. Key Insights:

- Determined major factors affecting Falcon 9 first-stage landing success.
- Identified geographic trends and variations in success rates.

2. Model Performance:

- SVM and KNN achieved 83.33% accuracy.
- Decision Tree achieved 94.44% accuracy.

3. Main Conclusions:

- Launch site and payload mass significantly influence landing outcomes.
 - Decision Tree demonstrated the highest predictive performance.
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INTRODUCTION

*** Project Background and Context**

- In this capstone project, our objective is to predict whether the first stage of the Falcon 9 rocket will successfully land. SpaceX is able to offer launch services at a much lower cost than many competitors, mainly because of its capability to reuse the rocket's first stage. By accurately forecasting landing outcomes, we can better estimate launch costs and generate strategic insights for organizations competing with SpaceX in the commercial launch market.

*** Key Questions We Aim to Address**

- What factors affect the successful landing of the Falcon 9 first stage?
 - How can machine learning techniques be used to accurately predict landing outcomes?
 - Which machine learning model provides the most reliable performance in predicting landing success?
-

DATA COLLECTION AND DATA WRANGLING METHODOLOGY

* Executive Summary

- This project applies a structured approach to predict Falcon 9 first-stage landing success, combining data collection, preprocessing, analysis, visualization, and machine learning modeling.

* Data Collection

- Launch data was obtained from the SpaceX API, including details such as launch dates, sites, payload mass, and mission outcomes.
- Steps: SpaceX API Request (Initiate API Request, Fetch Launch Data, Store Data Locally) → Web Scraping Wikipedia Extract HTML (Extract HTML Table, Parse with BeautifulSoup, Convert to DataFrame) → Data Integration (SpaceX API Data & Wikipedia Data, Merge Datasets, Final Integrated Data)

* Data Wrangling

- The dataset was cleaned, standardized, and enriched through feature extraction and engineering to improve model performance.

*Exploratory Data Analysis (EDA)

- Visualized success rates, payload distributions, and launch sites using Matplotlib and Seaborn.
- Applied SQL queries to extract insights and address key analytical questions.

https://github.com/haduongphan123/IBM_Learning/blob/main/1.%20ADSC-Data-collection-api.ipynb

DATA COLLECTION- SPACEX API

- **Step 1: Send an API Request**

- Use Python's requests library to establish a connection with the SpaceX API.
- Access the following endpoint: <https://api.spacexdata.com/v4/launches>.

- **Step 2: Process the API Response**

- Transform the returned JSON data into a Python dictionary.
- Retrieve the necessary information, including launch date, launch site, payload mass, rocket type, and mission outcome.

- **Step 3: Save the Data Locally**

- Organize the extracted information into a pandas DataFrame.
- Store the DataFrame locally to enable further analysis and processing.

https://github.com/haduongphan123/IBM_Learning/blob/main/1.%20ADSC-Data-collection-api.ipynb

DATA COLLECTION- WEB SCRAPPING

- **Step 1: Perform Web Scraping**
 - Utilize Python's requests library to retrieve the HTML content from the specified Wikipedia page.
 - Use the following URL as the data source: https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches.
- **Step 2: Analyze the HTML Structure**
 - Apply BeautifulSoup to parse and navigate the downloaded HTML content.
 - Identify and extract the table that contains the Falcon 9 launch history records.
- **Step 3: Create and Refine the DataFrame**
 - Transform the extracted table into a pandas DataFrame.
 - Clean, organize, and standardize the data to ensure accuracy and consistency for further analysis.

https://github.com/haduongphan123/IBM_Learning/blob/main/2.ADSC-WebScrapping.ipynb

EDA AND INTERACTIVE VISUAL ANALYTICS METHODOLOGY

* **Interactive Visual Analytics**

- Created interactive launch site and outcome maps using Folium.
 - Built a Plotly Dash dashboard with dynamic controls (dropdowns and sliders) to explore success rates and payload ranges.
 - https://github.com/haduongphan123/IBM_Learning/blob/main/ADSC-eda-sql-part2.ipynb
-

PREDICTIVE ANALYSIS METHODOLOGY

*** Predictive Modeling and Evaluation**

- Developed and assessed classification models, including Logistic Regression, SVM, KNN, and Decision Trees.
- Applied GridSearchCV for hyperparameter optimization.
- Compared model accuracy and selected the best-performing approach for landing prediction.
- https://github.com/haduongphan123/IBM_Learning/blob/main/ADSC_ML_Prediction_Part_4.jupyterlite.ipynb

EDA WITH SQL

Aggregate Queries

- Computed total launches, success/failure counts, and success rates by site and rocket type.

Join Queries

- Merged launch data with related tables (e.g., rocket details) for integrated analysis.

Filtering Queries

- Applied conditions to isolate specific outcomes, dates, or rocket configurations.

Sorting Queries

- Organized data by date or performance metrics to detect trends and anomalies.

Subqueries

- Used nested queries to derive advanced metrics, such as average payload mass per launch site.
 - https://github.com/haduongphan123/IBM_Learning/blob/main/ADSC-cda-sql-part2.ipynb
-

EDA WITH VISUALIZATION

Exploratory Data Analysis (EDA) focuses on visually examining and summarizing key characteristics of the dataset. Its objective is to understand data distribution, detect patterns, and explore relationships between variables.

1. Histograms

- Used to examine the distribution of numerical variables such as payload mass, flight number, and success rate.
- Help identify central tendency, spread, skewness, and outliers.

2. Bar Charts

- Applied to compare categorical variables, including launch outcomes across sites or rocket types.
- Provide clear comparisons of frequencies and proportions.

3. Line Charts

- Used to analyze trends over time, such as yearly launch success rates.
- Reveal performance patterns and temporal changes.

4. Scatter Plots

- Explore relationships between numerical variables, such as payload mass and landing success.
- Help identify correlations and dependencies.

5. Heatmaps

- Visualize correlation matrices among numerical variables.
- Assist in detecting strong relationships and multicollinearity.

6. Box Plots

- Display data distribution through quartiles.
- Highlight spread, skewness, and potential outliers across categories.

https://github.com/haduongphan123/IBM_Learning/blob/main/ADSC-eda-python-part3.ipynb

INTERACTIVE MAP RESULTS USING FOLIUM

Object

1. Markers:

- Added markers to display SpaceX launch sites on the map.
- Each marker represents a specific geographic launch location.
- Provides clear spatial reference of launch activities.

2. Circles:

- Added circles around launch sites to represent proximity zones.
- Visualize potential impact areas and operational boundaries.
- Provide a clear view of safety perimeters around launch locations.

3. Lines:

- Drew lines to connect launch sites with nearby or related locations.
- Visualize spatial relationships and dependencies between points of interest.
- Enhance understanding of geographic connections.

https://github.com/haduongphan123/IBM_Learning/blob/main/ADSC-folium-part5.ipynb

BUILD DASHBOARD WITH PLOTLY DASH

Pie Chart

- Illustrates the distribution of successful and failed launches.
- Provides a clear overview of overall mission success rates.
- Enables stakeholders to quickly assess performance at a glance.

Payload Scatter Plot

- Visualizes the relationship between payload mass and launch success.
- Highlights potential correlations between payload characteristics and mission outcomes.
- Supports data-driven decisions in payload planning and operational strategy.

Launch Site Dropdown

- Allows users to select specific launch sites for targeted analysis.
- Enables focused filtering by geographic location.
- Supports regional comparisons and deeper site-level insights.

Payload Range Slider

- Enables dynamic adjustment of payload mass ranges.
 - Supports interactive exploration of payload impact on mission success.
 - Provides deeper insights into payload-related performance trends.
-

PREDICTIVE ANALYSIS (CLASSIFICATION) RESULTS

1. Data Preparation

- Standardized features to ensure balanced model contribution.
- Split the dataset into training and testing sets for validation.

2. Model Selection

- Evaluated multiple classification models, including SVM, Decision Trees, and KNN.
- Selected algorithms appropriate for binary classification objectives.

3. Hyperparameter Optimization

- Applied GridSearchCV to identify optimal parameter settings.
- Tuned key parameters such as C (SVM), max_depth (Decision Trees), and n_neighbors (KNN).

4. Model Evaluation

- Used cross-validation to ensure model stability and generalization.
- Assessed performance using metrics such as accuracy, precision, recall, and F1-score.

5. Model Refinement

- Iteratively improved models based on validation insights.
- Adjusted hyperparameters to enhance predictive reliability.

6. Best Model Selection

- Selected the model with the highest test accuracy.
 - Compared training and testing performance to prevent overfitting and ensure practical applicability.
-

EDA WITH VISUALIZATION RESULTS

Flight Number vs. Launch Site

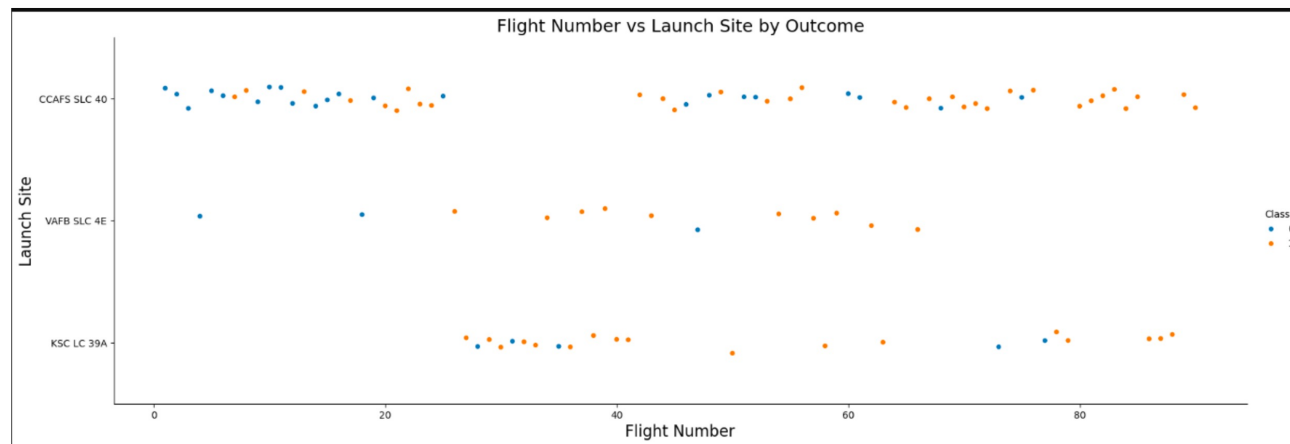
https://github.com/haduongphan123/IBM_Learning/blob/main/ADSC-eda-python-part3.ipynb

- **Mixed Landing Outcomes at Key Sites:**

Both CCAFS SLC 40 and KSC LC 39A show a combination of successful and unsuccessful landings, suggesting that landing performance depends on additional factors beyond the launch site itself.

- **Stable Launch Activity Across Flights:**

Launches are distributed across various flight numbers at all sites, indicating consistent operational activity over time without a clear upward or downward trend in landing success.

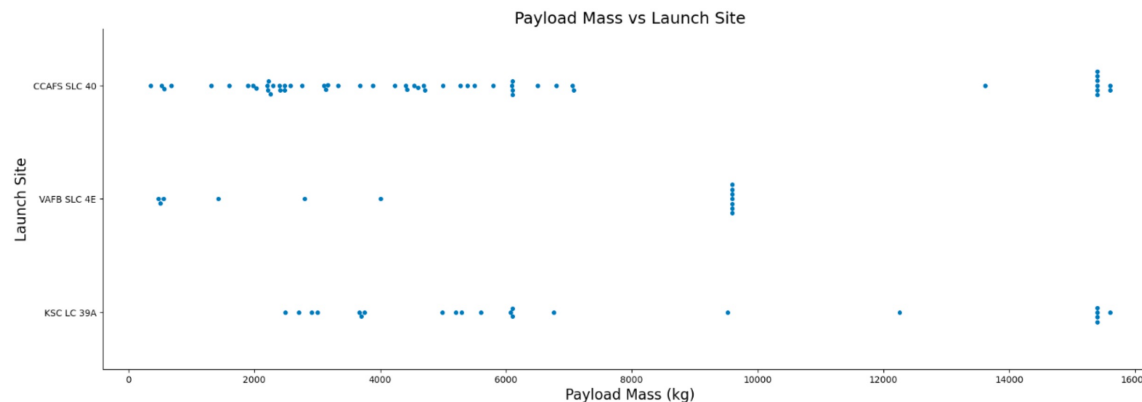


EDA WITH VISUALIZATION RESULTS

Payload vs. Launch Site

https://github.com/haduongphan123/IBM_Learning/blob/main/ADSC-eda-python-part3.ipynb

- **Payload Distribution by Launch Site:**
CCAFS SLC 40 primarily supports missions with payloads under 10,000 kg, while VAFB SLC 4E and KSC LC 39A handle a broader range of payload sizes, reflecting more diverse mission profiles.
- **High-Payload Missions at KSC LC 39A:**
KSC LC 39A is commonly used for heavier payload launches, including several missions exceeding 15,000 kg, highlighting its capacity for high-mass deployments.

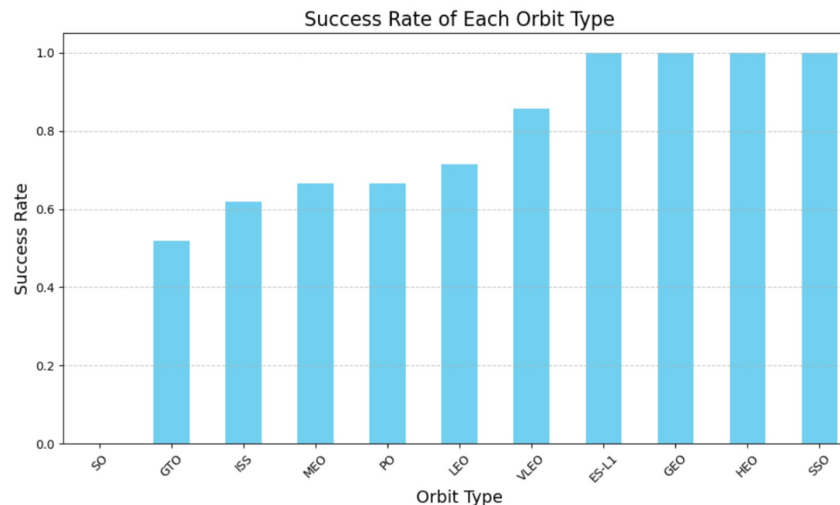


EDA WITH VISUALIZATION RESULTS

Success Rate vs. Orbit Type

https://github.com/haduongphan123/IBM_Learning/blob/main/ADSC-eda-python-part3.ipynb

- **Strong Success in Specific Orbits:**
Launches to VLEO, ES-L1, GEO, HEO, and SSO achieved a 100% landing success rate, indicating high reliability for these mission types.
- **Lower Performance in GTO Missions:**
GTO launches show a comparatively lower success rate, suggesting increased operational complexity or landing challenges for this orbit.

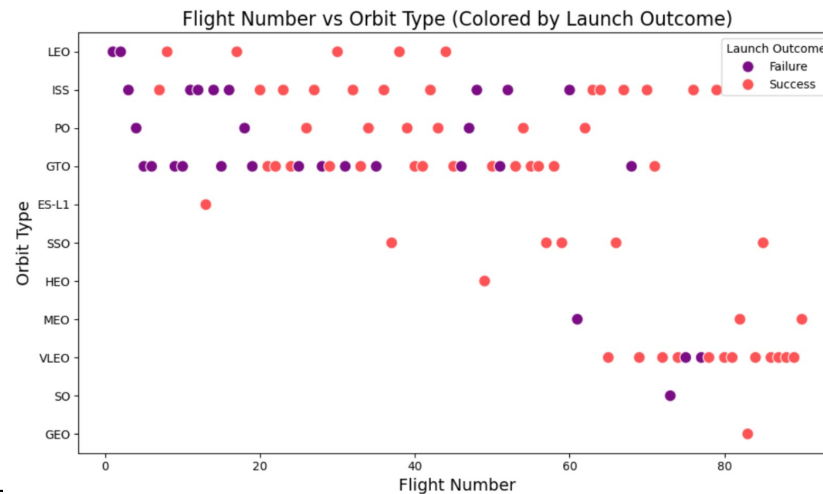


EDA WITH VISUALIZATION RESULTS

Flight Number vs. Orbit type

https://github.com/haduongphan123/IBM_Learning/blob/main/ADSC-eda-python-part3.ipynb

- **Improved Success Over Time:**
Falcon 9 landing success increases with higher flight numbers, suggesting that accumulated experience and continuous improvements enhance performance.
- **Enhanced Orbit Performance:**
While early GTO and ISS missions showed mixed results, more recent launches demonstrate higher success rates, reflecting progress in mission execution and operational efficiency.

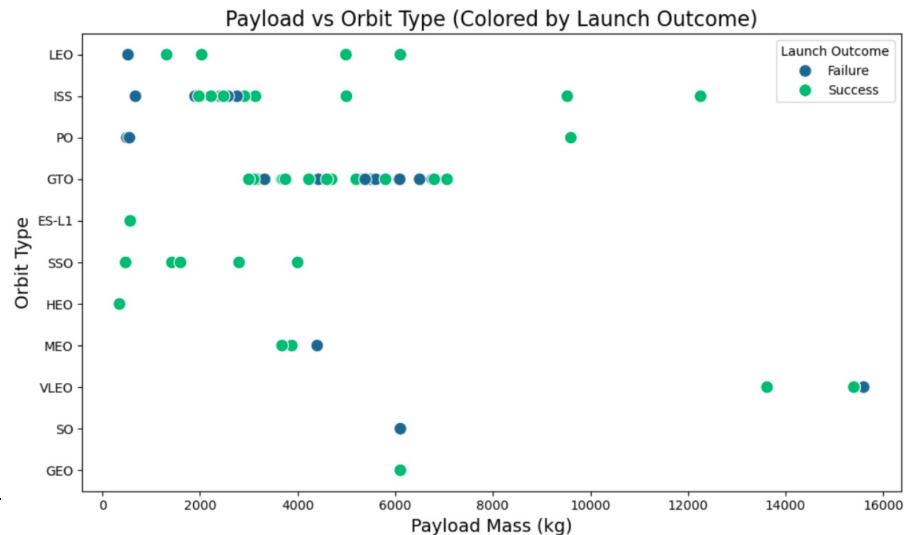


EDA WITH VISUALIZATION RESULTS

Payload vs. Orbit Type

https://github.com/haduongphan123/IBM_Learning/blob/main/ADSC-eda-python-part3.ipynb

- **Landing Success by Payload Mass:**
Successful landings are more common across all orbit types, particularly for payloads under 6,000 kg.
- **Challenges with Heavier Payloads:**
Missions carrying over 10,000 kg display mixed outcomes, suggesting greater operational complexity with higher payload masses.

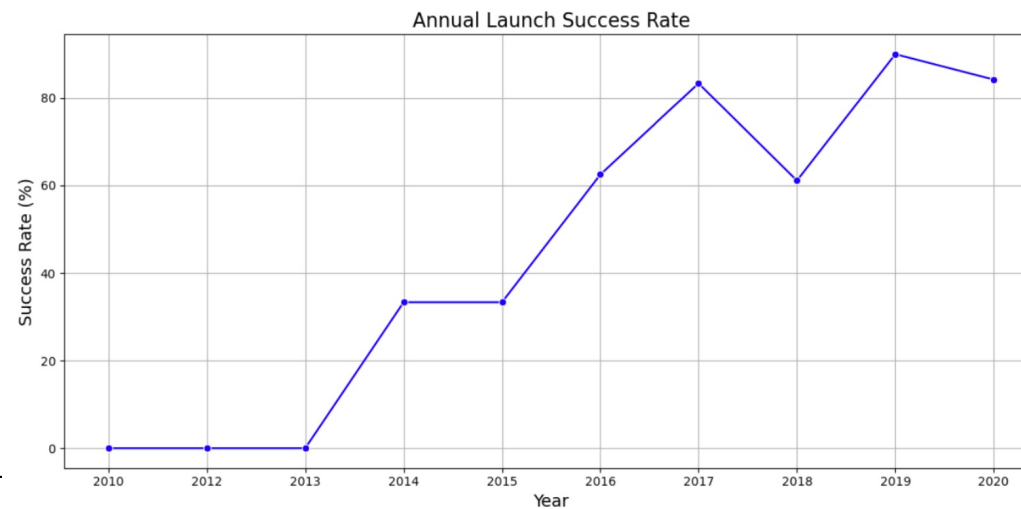


EDA WITH VISUALIZATION RESULTS

Launch Success Yearly Trend

https://github.com/haduongphan123/IBM_Learning/blob/main/ADSC-eda-python-part3.ipynb

- **Rising Annual Success Rate:**
The launch success rate has steadily improved since 2013, surpassing 80% by 2020.
- **Overall Reliability Growth:**
Although there was a temporary decline in 2018, the long-term trend reflects increasing reliability of Falcon 9 launches.



EDA WITH SQL RESULTS

All Launch Site Names

https://github.com/haduongphan123/IBM_Learning/blob/main/A/DSC-eda-sql-part2.ipynb

Task 1

Display the names of the unique launch sites in the space mission

```
[21]: %sql SELECT DISTINCT "Launch_Site" FROM SPACEXTABLE;
```

```
* sqlite:///my_data1.db
```

Done.

```
[21]: Launch_Site
```

```
CCAFS LC-40
```

```
VAFB SLC-4E
```

```
KSC LC-39A
```

```
CCAFS SLC-40
```

EDA WITH SQL RESULTS

Launch Site Names Begin with 'CCA'

Task 2

Display 5 records where launch sites begin with the string 'CCA'

```
[26]: %sql SELECT * FROM SPACEXTABLE WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5;
```

* sqlite:///my_data1.db
Done.

[26]:

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

https://github.com/haduongphan123/IBM_Learning/blob/main/ADSC-eda-sql-part2.ipynb

EDA WITH SQL RESULTS

Total Payload Mass

```
Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

[30]: %sql SELECT SUM("PAYLOAD_MASS_KG_") FROM SPACEXTABLE WHERE "Customer" = 'NASA (CRS)';
      * sqlite:///my_data1.db
      Done.
[30]: SUM(PAYLOAD_MASS_KG_)
      45596
```

https://github.com/haduongphan123/IBM_Learning/blob/main/ADSC-eda-sql-part2.ipynb

EDA WITH SQL RESULTS

Average Payload Mass by F9 v1.1

Task 4

Display average payload mass carried by booster version F9 v1.1

```
[34]: %sql SELECT AVG("PAYLOAD_MASS_KG_") FROM SPACEXTABLE WHERE "Booster_Version" = 'F9 v1.1';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
[34]: AVG(PAYLOAD_MASS_KG_)
```

```
2928.4
```

https://github.com/haduongphan123/IBM_Learning/blob/main/ADSC-eda-sql-part2.ipynb

EDA WITH SQL RESULTS

First Successful Ground Landing Date

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint: Use min function

```
[36]: %sql SELECT MIN("Date") FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (ground pad)';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
[36]: MIN(Date)
```

```
2015-12-22
```

https://github.com/haduongphan123/IBM_Learning/blob/main/ADSC-eda-sql-part2.ipynb

EDA WITH SQL RESULTS

Successful Drone Ship Landing with Payload between 4000 and 6000

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
[38]: %sql SELECT DISTINCT "Booster_Version" FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (drone ship)' AND "PAYLOAD_MASS_KG_" > 4000 AND "PAYLOAD_MASS_KG_" < 6000
```

```
* sqlite:///my_data1.db  
Done.
```

```
[38]: Booster_Version
```

```
F9 FT B1022
```

```
F9 FT B1026
```

```
F9 FT B1021.2
```

```
F9 FT B1031.2
```

https://github.com/haduongphan123/IBM_Learning/blob/main/ADSC-eda-sql-part2.ipynb

EDA WITH SQL RESULTS

Total Number of Successful and Failure Mission Outcomes

Task 7

List the total number of successful and failure mission outcomes

```
[40]: %sql SELECT "Mission_Outcome", COUNT(*) AS "Total" FROM SPACEXTABLE WHERE "Mission_Outcome" IN ('Success', 'Failure') GROUP BY "Mission_Outcome";
* sqlite:///my_data1.db
Done.
```

Mission_Outcome	Total
Success	98

https://github.com/haduongphan123/IBM_Learning/blob/main/ADSC-eda-sql-part2.ipynb

EDA WITH SQL RESULTS

Boosters Carried Maximum Payload

```
Task 8
List the names of the booster_versions which have carried the maximum payload mass. Use a subquery

[42]: %sql SELECT DISTINCT "Booster_Version" FROM SPACEXTABLE WHERE "PAYLOAD_MASS_KG_" = (SELECT MAX("PAYLOAD_MASS_KG_") FROM SPACEXTABLE);
* sqlite:///my_data1.db
Done.
[42]: 

| Booster_Version |
|-----------------|
| F9 B5 B1048.4   |
| F9 B5 B1049.4   |
| F9 B5 B1051.3   |
| F9 B5 B1056.4   |
| F9 B5 B1048.5   |
| F9 B5 B1051.4   |
| F9 B5 B1049.5   |
| F9 B5 B1060.2   |
| F9 B5 B1058.3   |
| F9 B5 B1051.6   |
| F9 B5 B1060.3   |
| F9 B5 B1049.7   |


```

https://github.com/haduongphan123/IBM_Learning/blob/main/ADSC-eda-sql-part2.ipynb

EDA WITH SQL RESULTS

2015 Launch Records

https://github.com/haduongphan123/IBM_Learning/blob/main/ADS-C-eda-sql-part2.ipynb

```
Task 9
List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.
Note: SQLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

[69]: %sql
SELECT
CASE
WHEN substr("Date", 6, 2) = '01' THEN 'January'
WHEN substr("Date", 6, 2) = '02' THEN 'February'
WHEN substr("Date", 6, 2) = '03' THEN 'March'
WHEN substr("Date", 6, 2) = '04' THEN 'April'
WHEN substr("Date", 6, 2) = '05' THEN 'May'
WHEN substr("Date", 6, 2) = '06' THEN 'June'
WHEN substr("Date", 6, 2) = '07' THEN 'July'
WHEN substr("Date", 6, 2) = '08' THEN 'August'
WHEN substr("Date", 6, 2) = '09' THEN 'September'
WHEN substr("Date", 6, 2) = '10' THEN 'October'
WHEN substr("Date", 6, 2) = '11' THEN 'November'
WHEN substr("Date", 6, 2) = '12' THEN 'December'
ELSE 'Unknown'
END AS "Month_Name",
"Mission_Outcome",
"Booster_Version",
"Launch_Site"
FROM
SPACEXTABLE
WHERE
substr("Date", 0, 5) = '2015';
* sqlite:///my_data1.db
Done.

[69]:
```

Month_Name	Mission_Outcome	Booster_Version	Launch_Site
January	Success	F9 v1.1 B1012	CCAFS LC-40
February	Success	F9 v1.1 B1013	CCAFS LC-40
March	Success	F9 v1.1 B1014	CCAFS LC-40
April	Success	F9 v1.1 B1015	CCAFS LC-40
April	Success	F9 v1.1 B1016	CCAFS LC-40
June	Failure (in flight)	F9 v1.1 B1018	CCAFS LC-40
December	Success	F9 FT B1019	CCAFS LC-40

EDA WITH SQL RESULTS

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

https://github.com/haduon-gphan123/IBM_Learning/blob/main/ADSC-eda-sql-part2.ipynb

```
Task 10
Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

1]: %%sql
SELECT
    "Landing_Outcome",
    COUNT(*) AS "Count"
FROM
    SPACEXTABLE
WHERE
    "Date" BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY
    "Landing_Outcome"
ORDER BY
    COUNT(*) DESC;

* sqlite:///my_data1.db
Done.

[81]:
```

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

INTERACTIVE MAP RESULTS USING FOLIUM

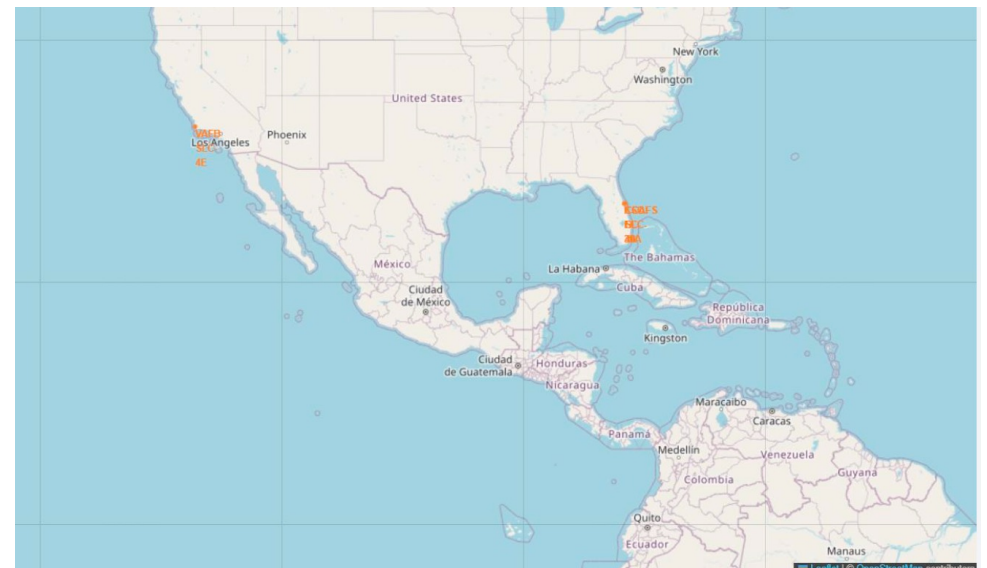
Mark all launch sites on a map

1. Proximity to the Equator:

Not all launch sites are close to the Equator. VAFB SLC-4E, located at latitude 34.63, is significantly farther from the Equator compared to the Florida launch sites.

2. Proximity to the Coast:

All launch sites are situated near coastal areas. Both the Florida sites (CCAFS LC-40 and KSC LC-39A) and VAFB SLC-4E in California are located close to the coastline.

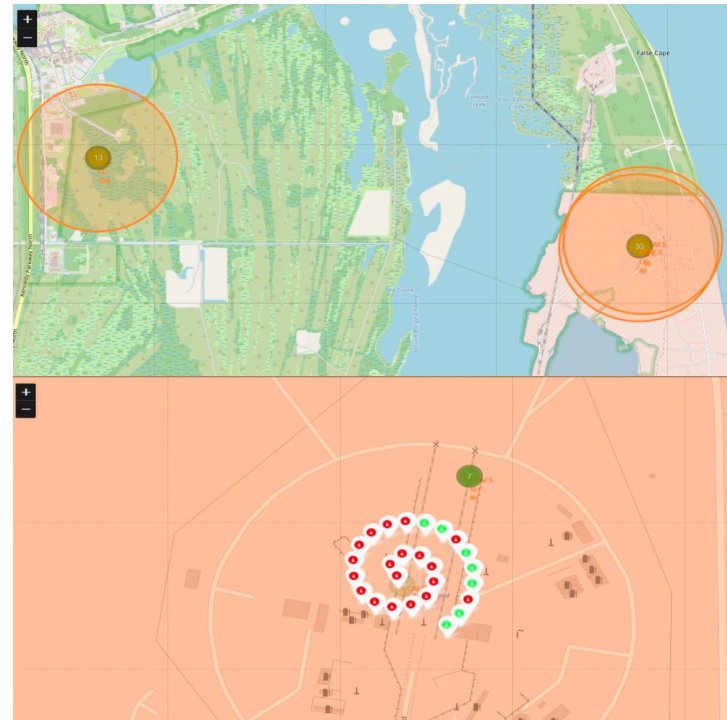


https://github.com/haduongphan123/IBM_Learning/blob/main/ADSC-folium-part5.ipynb

INTERACTIVE MAP RESULTS USING FOLIUM

Mark the success/failed launches for each site on the map

- Clustered markers improve the visualization and analysis of SpaceX launch data by organizing multiple locations more clearly. This approach reduces clutter and helps reveal patterns that may not be visible in a standard map view. Marker colors and pop-up details provide quick insights into launch characteristics and distribution.
- For instance, at CCAFS LC-40, 26 launches are displayed, including 19 red markers and 7 green markers. The color coding allows users to quickly distinguish between unsuccessful and successful launches, offering immediate visual insight into site performance.

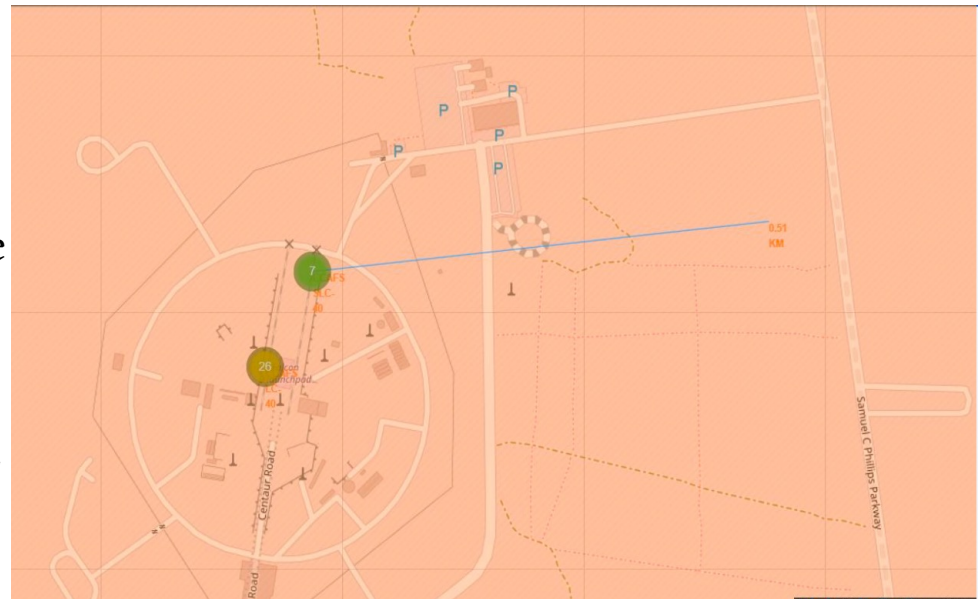


https://github.com/haduongphan123/IBM_Learning/blob/main/ADSC-folium-part5.ipynb

INTERACTIVE MAP RESULTS USING FOLIUM

Calculate the distances between a launch site to its proximities

This visualization illustrates the distance between the CCAFS SLC-40 launch site and the nearest coastline, approximately 0.51 kilometers. The PolyLine clearly displays the direct path, emphasizing the site's close coastal proximity. Such positioning supports over-water flight trajectories and enhances safety by minimizing risks to populated areas.

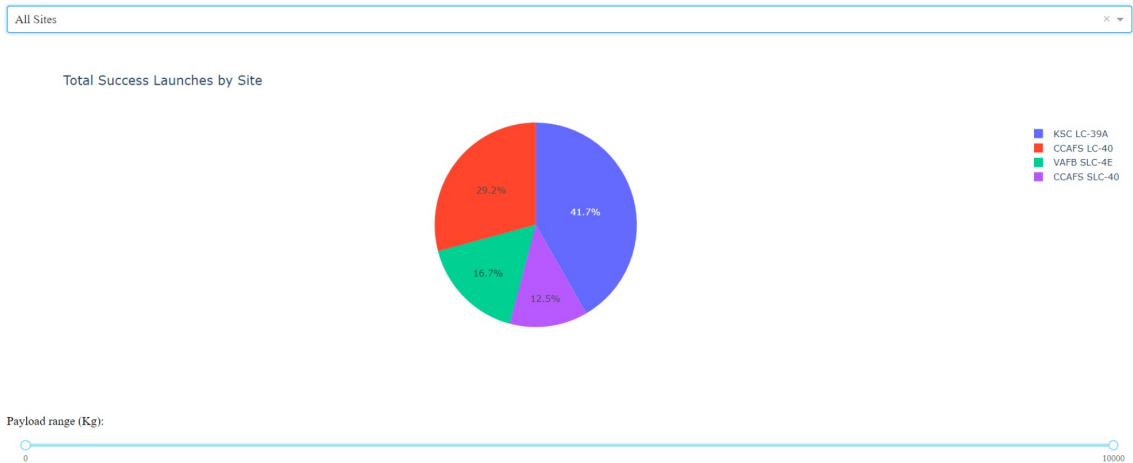


https://github.com/haduongphan123/IBM_Learning/blob/main/ADSC-folium-part5.ipynb

PLOTLY DASH DASHBOARD

Launch Success Count for all site

SpaceX Launch Records Dashboard



- CCAFS LC-40: 29.2%
 - CCAFS SLC-40: 12.5%
 - VAFB SLC-4E: 16.7%
 - KSC LC-39A: 41.7%
 - Among all sites, KSC LC-39A accounts for the largest portion of successful launches (41.7%), demonstrating its high operational effectiveness and reliability.
-

PLOTLY DASH DASHBOARD

Pie chart for the launch site with highest launch success ratio

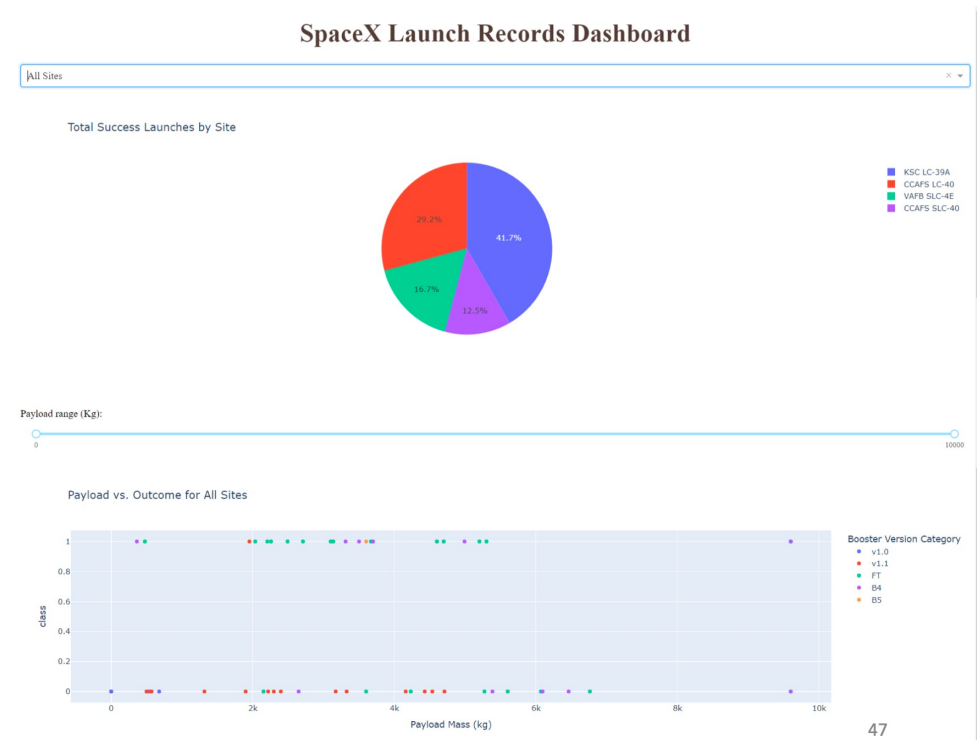
- The strong share of successful missions from KSC LC-39A emphasizes its reliability and operational efficiency.
- **KSC LC-39A Performance:**
 - Successful launches (Class 1): 76.9%
 - Unsuccessful launches (Class 0): 23.1%
- With a 76.9% success rate, KSC LC-39A stands out as one of the most dependable launch sites



PLOTLY DASH DASHBOARD

Insights from SpaceX Launch Data Dashboard

- **Launch Site Success Rates:**
 - **CCAFSLC-40** has the highest success rate
- with 43.7% of successful launches.
- ○ This suggests that **CCAFS LC-40** is the most reliable launch site among the ones analyzed.
- ○ Other sites like **KSC LC-39A**, **VAFB SLC- 4E**, and **CCAFS SLC-40** have lower success rates, indicating variability in launch success across different sites.
- **Booster Version Performance:**
 - **Booster version “FT”** appears to be the most frequently used and has a high success rate across various payload masses.
 - **Booster version “v1.0”** has fewer launches and may require further analysis to understand its performance.
 - Overall, booster versions do not show a clear trend that higher payload masses correlate with lower success rates.

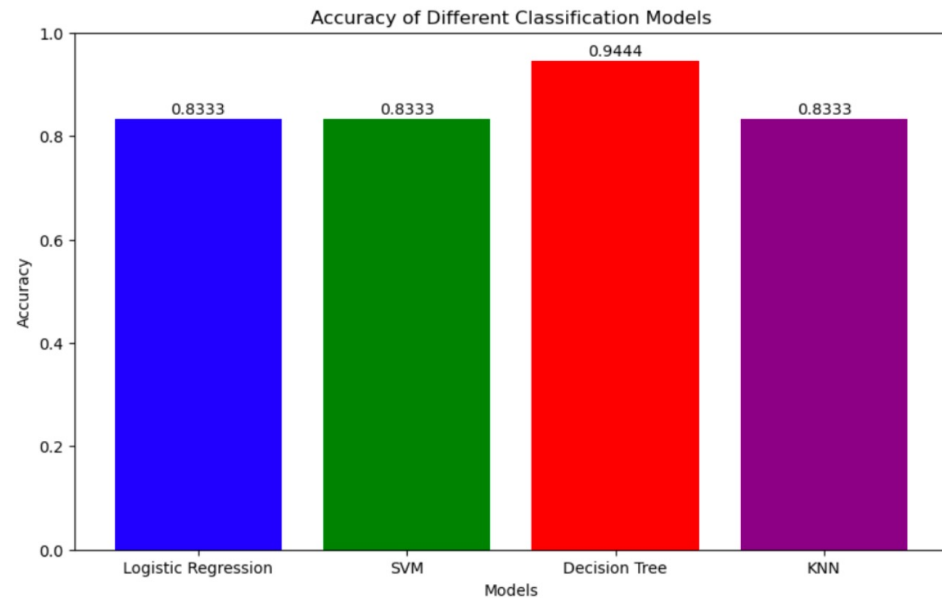


PREDICTIVE ANALYSIS (CLASSIFICATION)

Accuracy

- The Decision Tree model achieved the highest test accuracy at **94.44%**, outperforming the other models. In comparison, Logistic Regression, SVM, and KNN each achieved **83.33%** accuracy.
- These results indicate that the Decision Tree model is the most suitable approach for this dataset.

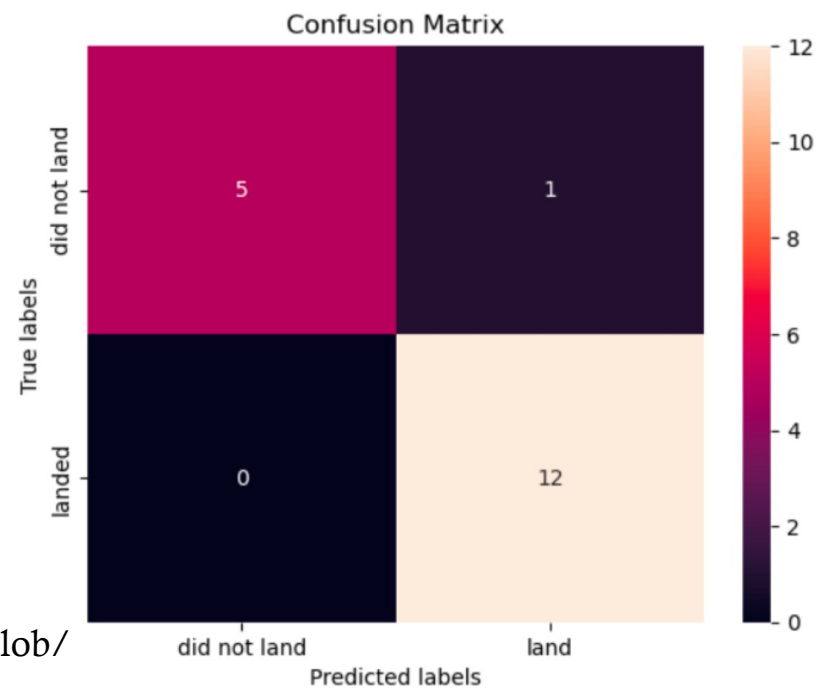
https://github.com/haduongphan123/IBM_Learning/blob/main/ADSC_ML_Prediction_Part_4.jupyterlite.ipynb



PREDICTIVE ANALYSIS (CLASSIFICATION)

Confusion matrix

- **High Accuracy:** The model achieved a high accuracy score of 94.44%, with a significant number of true positives and true negatives, demonstrating its effectiveness in predicting Falcon 9 first stage landings.
- **No False Negatives:** The absence of false negatives indicates that the model reliably predicts successful landings. This is crucial for ensuring readiness and safety in aerospace operations, as every actual successful landing was accurately identified.
- **Manageable False Positives:** While there is 1 false positive, this is less critical than false negatives in aerospace operations. Over- preparation (due to false positives) is more manageable than under-preparation, making the model's performance highly acceptable for practical applications.
- **Balanced Performance:** The model shows a balanced performance with a slight bias towards predicting successful landings. This aligns well with practical needs in the aerospace industry, where ensuring successful landings is of paramount importance for cost estimation and planning.



https://github.com/haduongphan123/IBM_Learning/blob/main/ADSC_ML_Prediction_Part_4.jupyterlite.ipynb

CONCLUSION

- **Point 1:**
Our findings indicate that **CCAFS LC-40** achieved the highest launch success rate, contributing 43.7% of successful missions. This suggests that the site may benefit from more favorable operational conditions or established procedures.
 - **Point 2:**
The scatter plot analysis revealed that the **FT booster version** maintains a consistently high success rate across different payload masses, highlighting its reliability and operational stability.
 - **Point 3:**
No strong correlation was identified between higher payload mass and lower launch success, implying that other factors—such as launch site conditions and booster version—have a greater influence on outcomes.
 - **Point 4:**
Interactive visualizations using Folium and Plotly Dash enhanced our understanding of geographic and operational trends, supporting more informed, data-driven decision-making.
 - **Conclusion:**
Overall, the predictive modeling and visual analysis provided meaningful insights into the factors affecting launch success and established a solid framework for future evaluation and strategic planning in the aerospace sector.
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